

CUSTOMER LIFETIME VALUE PREDICTION

```
from google.colab import files
```

```
files.upload()
```

```
import zipfile
```

```
import os
```

```
# Path to uploaded zip file (update if needed)
```

```
zip_path = "/content/archive (1).zip"
```

```
extract_path = "/content/final_data"
```

```
# Extract the zip
```

```
with zipfile.ZipFile(zip_path, 'r') as zip_ref:
```

```
    zip_ref.extractall(extract_path)
```

```
# List extracted files
```

```
os.listdir(extract_path)
```

```
↔ ['online_retail_II.xlsx']
```

```
# Install dependencies
```

```
!pip install openpyxl seaborn scikit-learn --quiet
```

```
# Import libraries
```

```
import pandas as pd
```

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

```
from sklearn.ensemble import RandomForestRegressor
```

```
from sklearn.model_selection import train_test_split
```

```
from sklearn.metrics import mean_absolute_error, r2_score
```

```
# Load the dataset
```

```
file_path = "/content/final_data/online_retail_II.xlsx"
```

```
df = pd.read_excel(file_path, sheet_name='Year 2010-2011')
```

```
# Clean the data
```

```
# Remove rows with missing Customer ID
```

```
df = df.dropna(subset=['Customer ID'])
```

```
# Remove cancelled orders and make a copy to avoid SettingWithCopyWarning
```

```
df = df[~df['Invoice'].astype(str).str.startswith('C')].copy()
```

```
# Add new calculated columns
```

```
df['TotalPrice'] = df['Quantity'] * df['Price']
```

```
df['InvoiceDate'] = pd.to_datetime(df['InvoiceDate'])
```

```
# RFM Feature Engineering
```

```
snapshot_date = df['InvoiceDate'].max() + pd.DateOffset(days=1)
```

```
rfm = df.groupby('Customer ID').agg({  
    'InvoiceDate': lambda x: (snapshot_date - x.max()).days,  
    'Invoice': 'nunique',  
    'TotalPrice': 'sum'  
}).rename(columns={  
    'InvoiceDate': 'Recency',  
    'Invoice': 'Frequency',  
    'TotalPrice': 'Monetary'  
}).reset_index()
```

```
# Create target: Future Value (last 6 months)
```

```
cutoff = df['InvoiceDate'].max() - pd.DateOffset(months=6)
```

```
future = df[df['InvoiceDate'] > cutoff].groupby('Customer ID')['TotalPrice'].sum()
```

```
rfm = rfm.merge(future.rename('FutureValue'), on='Customer ID',  
how='left').fillna(0)
```

```
# Split and train model
```

```
X = rfm[['Recency', 'Frequency', 'Monetary']]
```

```
y = rfm['FutureValue']
```

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,  
random_state=42)
```

```
model = RandomForestRegressor(random_state=42)
```

```
model.fit(X_train, y_train)
```

```
# Predict and evaluate
```

```
y_pred = model.predict(X_test)
```

```
mae = mean_absolute_error(y_test, y_pred)
```

```
r2 = r2_score(y_test, y_pred)
```

```
# 📊 Visualizations
```

```
plt.figure(figsize=(15, 5))
```

```
plt.subplot(1, 3, 1)
```

```
sns.histplot(rfm['Recency'], kde=True, color='skyblue')
```

```
plt.title("Recency Distribution")
```

```
plt.subplot(1, 3, 2)
```

```
sns.histplot(rfm['Frequency'], kde=True, color='orange')
```

```
plt.title("Frequency Distribution")
```

```
plt.subplot(1, 3, 3)
```

```
sns.histplot(rfm['Monetary'], kde=True, color='green')
```

```
plt.title("Monetary Distribution")
```

```
plt.tight_layout()
```

```
plt.show()
```

```
# Heatmap of correlations
```

```
plt.figure(figsize=(6,4))

sns.heatmap(rfm[['Recency', 'Frequency', 'Monetary', 'FutureValue']].corr(),
annot=True, cmap='coolwarm')

plt.title("Correlation Heatmap")

plt.show()
```

Feature Importance

```
feat_importance = pd.Series(model.feature_importances_, index=X.columns)

feat_importance.sort_values().plot(kind='barh', title='Feature Importance',
color='teal')

plt.xlabel("Importance")

plt.show()
```

📄 Print Model Evaluation

```
print("📈 Model Evaluation:")

print(f"Mean Absolute Error: ₹{mae:.2f}")

print(f"R2 Score: {r2:.2f}")
```

📊 Visual Business Insights

Top 10 Countries by Revenue

```
plt.figure(figsize=(10, 5))

top_countries =
df.groupby('Country')['TotalPrice'].sum().sort_values(ascending=False).head(10
)

sns.barplot(x=top_countries.values, y=top_countries.index, color='skyblue')

plt.title("Top 10 Countries by Total Revenue")
```

```
plt.xlabel("Revenue")
```

```
plt.ylabel("Country")
```

```
plt.tight_layout()
```

```
plt.show()
```

```
# Average Order Value per Customer
```

```
customer_order_value = df.groupby('Customer ID')['TotalPrice'].sum() /  
df.groupby('Customer ID')['Invoice'].nunique()
```

```
plt.figure(figsize=(8, 4))
```

```
sns.histplot(customer_order_value, bins=30, color='mediumorchid')
```

```
plt.title("Average Order Value per Customer")
```

```
plt.xlabel("Order Value")
```

```
plt.ylabel("Number of Customers")
```

```
plt.tight_layout()
```

```
plt.show()
```

```
# Monthly Revenue Trend
```

```
df['Month'] = df['InvoiceDate'].dt.to_period('M')
```

```
monthly_revenue = df.groupby('Month')['TotalPrice'].sum()
```

```
monthly_revenue.index = monthly_revenue.index.astype(str)
```

```
plt.figure(figsize=(12, 5))
```

```
monthly_revenue.plot(marker='o', color='darkblue')
```

```
plt.title("Monthly Revenue Trend")
```

```
plt.xlabel("Month")
```

```
plt.ylabel("Revenue")
```

```
plt.grid(True)
```

```
plt.xticks(rotation=45)
```

```
plt.tight_layout()
```

```
plt.show()
```

```
# Customer Segmentation by Predicted Future Value
```

```
rfm['CustomerSegment'] = pd.qcut(rfm['FutureValue'], 4, labels=['Low', 'Mid',  
'High', 'Top'])
```

```
plt.figure(figsize=(7, 4))
```

```
sns.countplot(data=rfm, x='CustomerSegment', order=['Low', 'Mid', 'High',  
'Top'], palette='Set2')
```

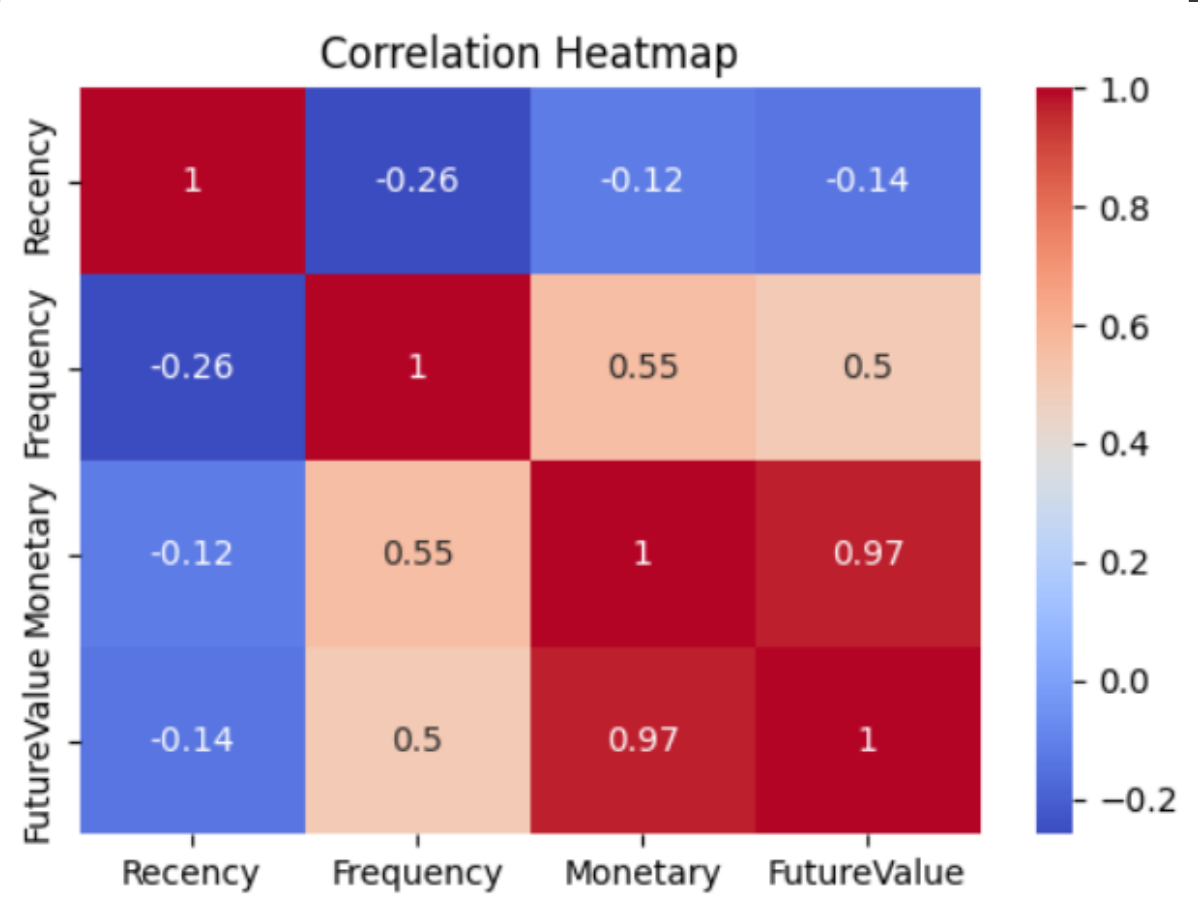
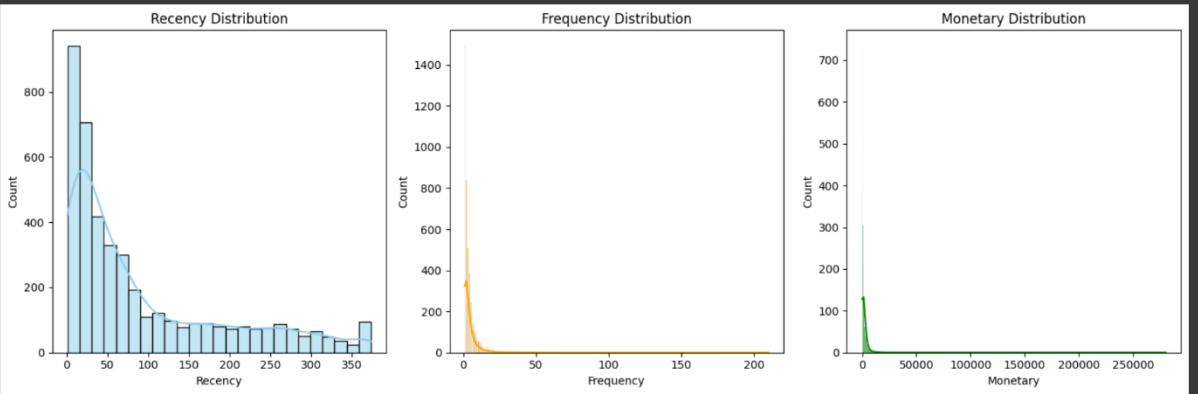
```
plt.title("Customer Segments Based on Predicted Future Value")
```

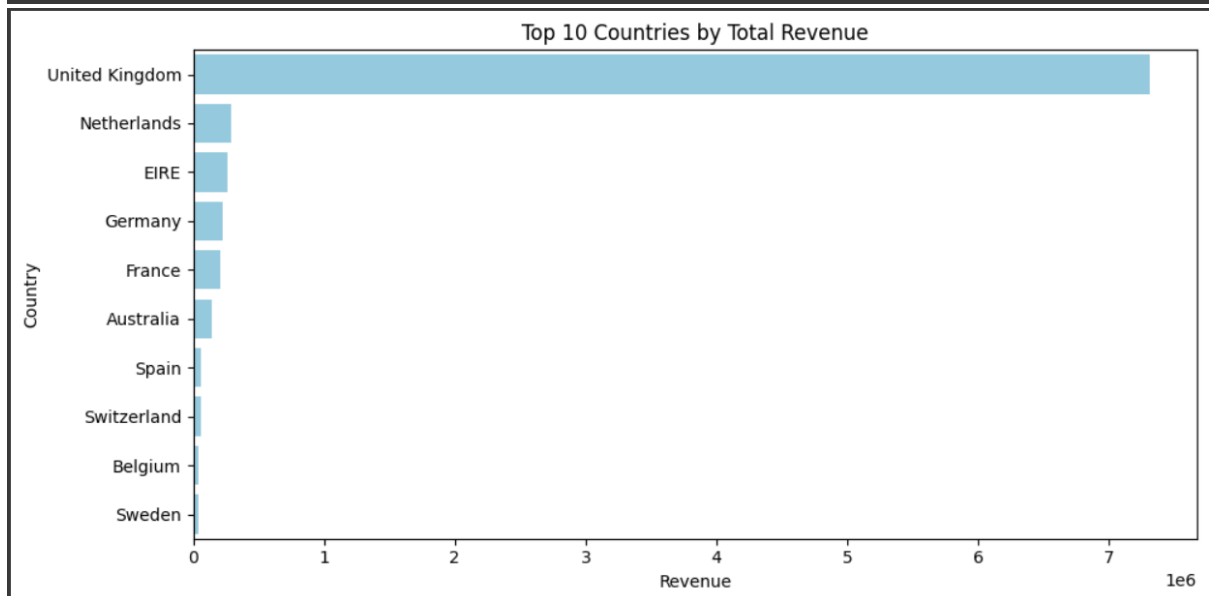
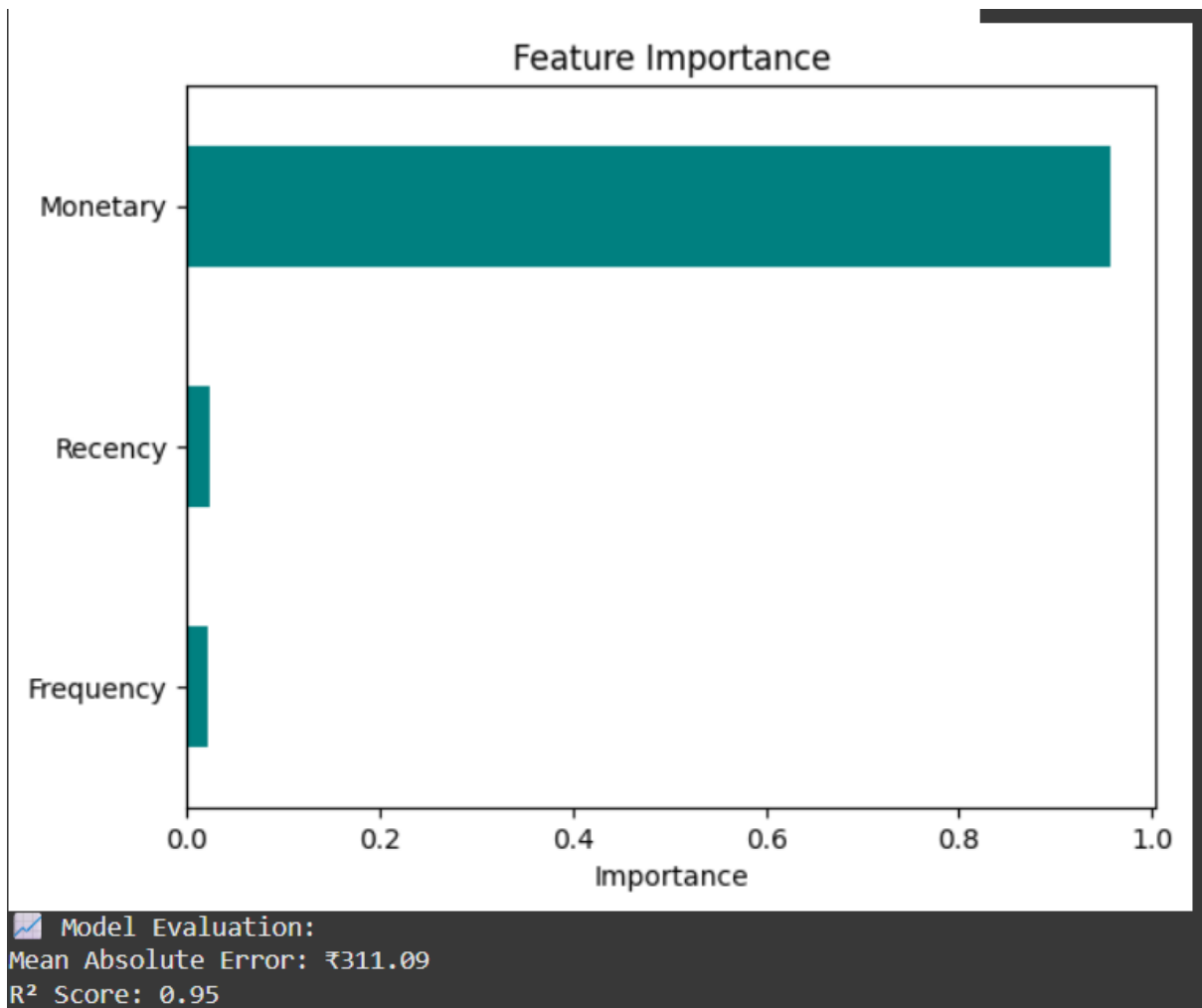
```
plt.xlabel("Customer Segment")
```

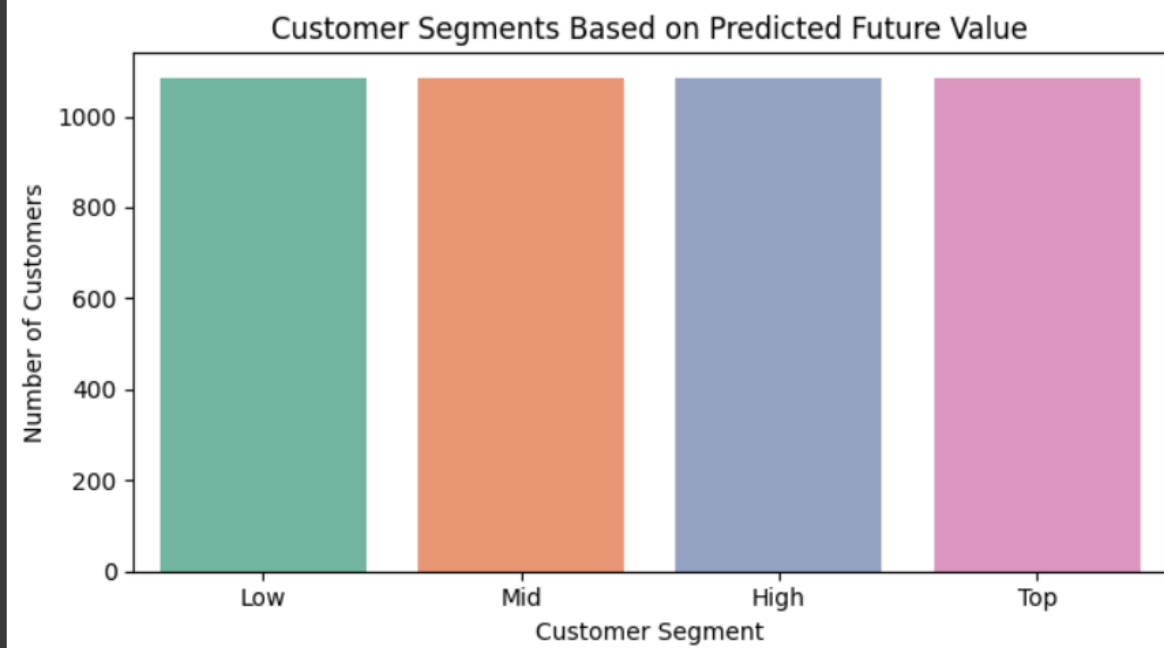
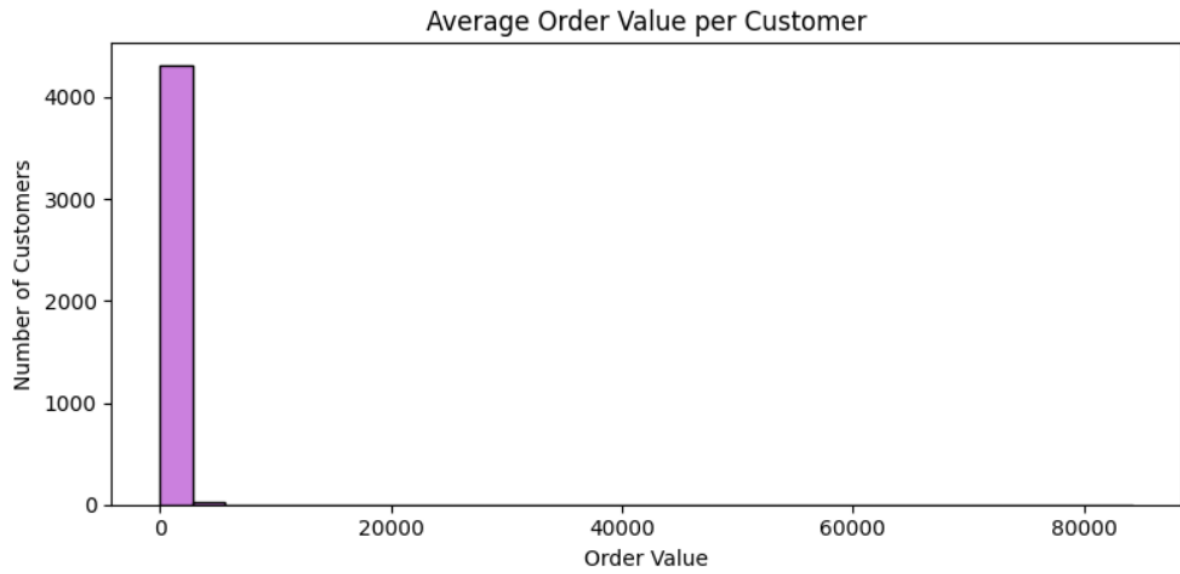
```
plt.ylabel("Number of Customers")
```

```
plt.tight_layout()
```

```
plt.show()
```







CONCLUSION

In this project, we analyzed customer transaction data from an online retail store to predict **Customer Lifetime Value (CLV)** — the future monetary worth of a customer to the business.

- **Data cleaned** by removing canceled invoices and missing customer IDs
- **RFM features** (Recency, Frequency, Monetary) used to describe customer behavior
- **Target variable** defined as total spending in the last 6 months
- **Random Forest Regressor** trained to predict Customer Lifetime Value (CLV)
- Model evaluated using **Mean Absolute Error** and **R² Score**
- **Top revenue-generating countries** identified
- **Monthly revenue trends** show seasonal buying behavior
- **Customer segments** (Low, Mid, High, Top) created using predicted CLV
- Most customers had **low-frequency, low-value** purchases — opportunity for retention strategies